Portlogi: A Departure-Aware Digital Twin for Container Yard Placement and Truck Booking at Indian Ports

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***Abstract*—A container that sits on top of one leaving earlier forces an extra lift, called a rehandle, and every rehandle costs vehicle time and keeps a truck waiting at the gate. Indian ports have it worse than most, since boxes dwell for a long time and the pickup time is usually unknown at the moment a box is placed. We describe Portlogi, a simulated smart yard for V. O. Chidambaranar Port, Thoothukudi. A box is admitted when its RFID card is read. A slot is chosen using hard safety rules plus a five-part score, and a fleet of self-driving vehicles moves the box without ever sharing a cell. Businesses can also book a truck, which tells the yard the pickup time. On a small yard the allocator cut rehandles by 69.6% against first-available placement (6.7 down to 2.0 per 30 boxes over 100 runs), and by about 31% against a rule that already uses departure times. A second experiment shows that a pickup time lowers rehandles only if the yard has it when the box is placed. A later booking still removes truck waiting, though. Across 200 random yards no safety rule was broken. The site runs on AWS and scored A+ on two public security scans. Everything reported here comes from simulation, and the RFID gate hardware is planned but not yet built.**

***Index Terms*—container yard, rehandling, digital twin, smart port, truck booking, RFID**

I. Introduction

Ports are the doors of a city's economy. Containers come off a ship, sit in a yard, and leave by truck or rail, and to save space the yard stacks them. That saving has a price: when the box you need is buried, whatever lies on top has to be lifted away first, and this extra lift is the rehandle.

Dwell time in Indian yards is long. A report on the first quarter of 2024 puts the median export dwell time at more than four days, up from about three and a half a year before [8]. Boxes that stay longer fill the yard, and a fuller yard produces more rehandles. On top of that, an import box has to be placed before anyone knows when its truck will arrive, so the best placement is at some level a guess.

The Tamil Nadu Smart Cities project covers Thoothukudi, home to V. O. Chidambaranar (VOC) Port, which handled 734,274 TEU in 2022-23 [10]. We picked it as the setting for a smart yard that doubles as a service for businesses. The port authority would run a booking desk that shipping lines and hauliers use directly, which is the government-to-business (G2B) idea.

Our objectives were (1) to place each box where it is unlikely to block another, (2) to move boxes with several vehicles and no collisions, (3) to let a business book a truck so that the yard learns the pickup time, (4) to record the reason behind every decision so any run can be replayed exactly, and (5) to offer the whole system over a secure web link.

TABLE I
What each line of work covers

Work	Stacking by time	Uncertain time	Traffic control	Booking desk
Dekker et al. [1]	yes	no	no	no
Ku and Arthanari [2]	yes	windows	no	no
Galle et al. [4]	yes	yes	no	no
Möhring et al. [5]	no	no	yes	no
Kim et al. [6]	no	no	yes	no
This work	yes	yes	yes	yes

Contribution. We built a working simulator containing all of these parts and measured it. We also ran an experiment on a question we could not find answered elsewhere: how much is a pickup time worth, and by when does it have to arrive?

II. Literature Review

Stacking with departure information is a well-studied topic. Dekker et al. compared stacking policies by simulation and found that using departure times lowers rehandles [1]. Caserta et al. proved that the blocks relocation problem is hard and offered both exact and rule-based methods [3]. Ku and Arthanari handled the case where departures arrive as time windows, for instance from a truck appointment system, and built a model with a fast heuristic [2]. Galle et al. moved on to the stochastic version, in which the retrieval order itself is uncertain [4].

On the traffic side, Möhring et al. describe conflict-free routing of automated guided vehicles (AGVs) through time windows, and they tested it at a Hamburg terminal [5]. Kim, Jeon and Ryu look at how AGV deadlocks arise and how to prevent them [6]. A vendor case study reports a 25% drop in rehandles after one terminal adopted a yard-planning tool [7], and in India the Logistics Data Bank follows containers using RFID tags [9].

Gap. Among the papers we read, none joins gate admission, placement, vehicle traffic and a business booking desk in one open system with a replayable record of each decision. Nor did we find any measurement of how the arrival time of a booking changes the outcome. Table I summarizes this, though it comes from the papers we happened to read and is not a full survey.

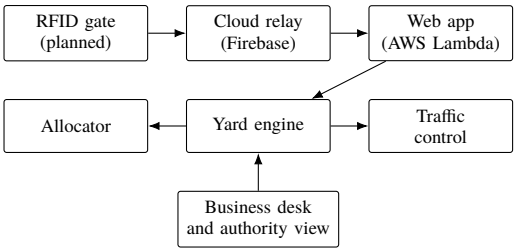


Fig. 1. System plan. The gate hardware is planned. The other parts are built.

III. Proposed Methodology

Fig. 1 gives the overall plan. When a card is read at the gate, the yard engine creates a container record and the allocator chooses a slot. A vehicle is then dispatched, a path is planned and the move is carried out. If the box is an import, a business can book a truck for it. That booking fixes the pickup time, and the yard brings the box to the gate ahead of the truck's due time.

A. Placement

Every free slot is first tested against hard rules. A reefer needs a powered slot, a heavy box may not rest on a lighter one, and a hazardous box may not touch another hazardous box. The slots that survive get a score, and the lowest score wins:

$$C(s) = 40\,r + 15\,t + 15\,c + 12\,p + 18\,a \qquad (1)$$

Here r is rehandle risk (how much the box would sit on boxes that leave earlier), t is the driving distance from the gate, c is how well the box groups with others bound for the same destination, p is how far the yard would spread into fresh stacks, and a is how much the slot would block the aisles. The weights can be changed by the user. When no legal slot exists, the box waits on a transfer pad until one opens.

B. Traffic

Vehicles reserve cells for time windows along their paths, so no two ever share a cell. Travel time for each segment is proportional to its real distance, which keeps every vehicle at a steady speed. If a ring of vehicles ends up waiting on one another, the system picks one by priority and moves it to a free cell until the ring clears.

C. Booking

An import box has no fixed pickup time, so we assume it leaves about 180 minutes after arrival and flag that assumption as a guess. Once a business books a haulier and a window, the booking replaces the guess as the real pickup time. The yard then digs the box out if needed and has it at the gate before the window opens. The desk also shows a quote that includes the cost of yard handling.

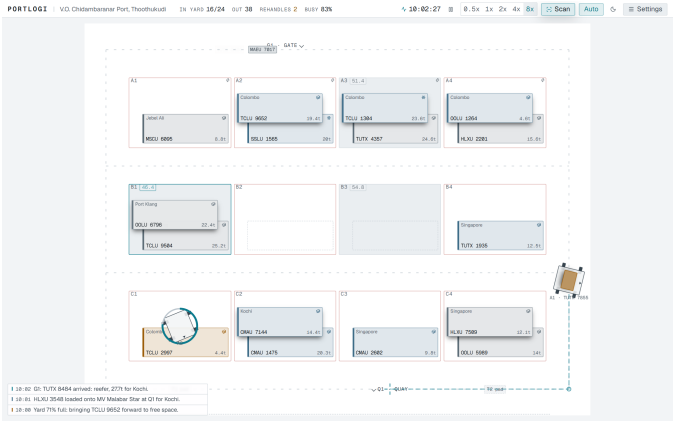


Fig. 2. The yard view during a run. Boxes show their destination and weight. A vehicle is moving a box.

IV. System Design and Implementation

Software. The engine is written in TypeScript and is fully deterministic. Time advances in fixed ticks of 0.01 simulated minutes and random numbers come from a seeded generator, so a given seed always produces the same result. Each user action is stored as a command in a journal, which lets us replay a run exactly. During simulation the engine checks that no two vehicles come closer than a set distance, that every stack is legal, and that no box is lost or duplicated. The interface is built with React and runs in the browser, while the heavy comparisons run in a web worker.

Hardware (planned). An ESP32 board with an RFID reader will push the card number to a cloud database. The database rules allow a gate device to add scan records but nothing else, reading requires a sign-in, and public sign-up is switched off. The web page then admits the box. This code is written, but we have not tried it on real hardware.

Hosting. The site runs on AWS Lambda behind a secure web link, with response headers that set a strict content security policy and other protections. A script checks the live site against 31 rules, and before each deploy another check confirms that only free-tier services are in use.

Ports. The landing page shows an OpenStreetMap map [11] with 17 Indian ports. Clicking one opens an illustrative yard with its own size, stack height, vehicles and settings. These yards are invented for the demo, since we could not find the real layouts.

V. Results and Discussion

Every result below comes from the simulator. Unless noted otherwise, the yard has 3 rows, 4 columns, 2 tiers and 2 vehicles, and values are the mean $\pm$ 95% confidence interval.

A. Placement quality

We ran 100 seeds of 30 arrivals each, giving both methods the same containers (Table II). Rehandles fell by 69.6%. The allocator did better in 96 runs, tied in 2 and did worse in 2, and no safety rule was broken in any of the 200 runs.

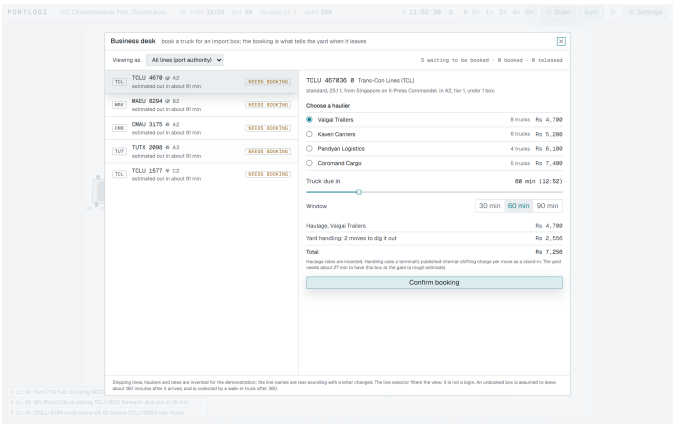


Fig. 3. The business desk. A shipping line picks a haulier and a window and sees a quote.

TABLE II
ALLOCATOR AGAINST FIRST AVAILABLE PLACEMENT (100 SEEDS)

Measure	First available	Allocator
Rehandles per run	6.7 ± 0.4	2.0 ± 0.3
Grid cells driven	917 ± 10	864 ± 9
Time to clear (min)	355 ± 7	345 ± 6
Average dwell (min)	112 ± 2	108 ± 2

First-available is a weak baseline, so we also compared against a rule that uses departure time alone, which is the rehandle term by itself. Over 60 seeds that rule averaged 3.12 rehandles per run against 2.15 for the full allocator, about 31% fewer. In other words the simple departure-time rule delivers roughly 80% of the total gain, and the other four terms add a smaller but steady improvement on top.

B. Ablation

In Table III we switched off one term at a time over 40 seeds. Dropping the rehandle term hurts most: rehandles climb to 7.1, no better than first-available at 7.0. The remaining terms matter less, and of those, aisle access is the one whose removal costs the most.

C. Bigger yards

Table IV covers four sizes with 12 seeds each, and the gap widens as the yard grows. At 60 positions the allocator needed 7.1 rehandles while first-available needed 56.7. Simulating a 60-position run takes about 1.6 seconds.

D. Fleet size

On the small yard, a single vehicle needed 371 minutes to clear the yard and left gate waits near 12 minutes. Two vehicles brought that down to 337 minutes and 5.2 minutes. A third added little (334 minutes and 3.7 minutes) and sat idle most of the time, with utilisation of 0.27.

TABLE III
ONE TERM REMOVED AT A TIME (40 SEEDS)

Setting	Rehandles	Time to clear (min)
All terms	2.0 ± 0.4	342 ± 9
Rehandle term only	2.9 ± 0.5	355 ± 11
No rehandle term	7.1 ± 0.6	346 ± 10
No travel	1.9 ± 0.5	343 ± 9
No cluster	2.2 ± 0.4	342 ± 10
No spread	2.1 ± 0.4	344 ± 9
No aisle access	2.4 ± 0.4	348 ± 11
First available	7.0 ± 0.5	353 ± 11

TABLE IV
SCALING WITH YARD SIZE (12 SEEDS)

Layout	Vehicles	Allocator	First avail.
3x4 (12)	2	0.9 ± 0.4	5.3 ± 1.2
4x6 (24)	4	2.8 ± 0.9	22.8 ± 2.2
5x8 (40)	5	5.7 ± 1.7	36.7 ± 3.4
6x10 (60)	6	7.1 ± 1.6	56.7 ± 3.1

E. What is a pickup time worth?

This experiment uses 20 import boxes and 32 seeds. Each box has a hidden true pickup time drawn between 1 and 5 hours after arrival, and we varied what the yard knows and when it learns it. The 1 to 5 hour range was our own choice, so the size of the effect is only illustrative and the pattern is what matters.

Two results stand out. First, a booking made anywhere from 20 minutes to 2 hours before pickup does not reduce rehandles (4.0 and 3.8, against 4.1 with no information), because the box is already stacked by then. Only a time known at arrival brings the number down, to 1.3. A good estimate, within 15 minutes, also helps (1.7) even without a firm booking. Second, any booking at all removes nearly all truck waiting, from about 10 minutes to almost nothing, since the yard has time to dig the box out and stage it. For a port authority the practical lesson is to ask for the booking when the box is discharged and not when the truck is about to leave.

F. Safety, repeatability and tests

We generated 200 random yards (1 to 6 rows, 2 to 10 columns, 1 to 4 tiers, 1 to 6 vehicles, with a closed slot in 40% of them). Each ran for 1000 simulated minutes with a vessel surge, a vehicle breakdown and a reefer storm. We saw no collision, no illegal stack, no lost or duplicated box, and no stalled yard. Ten seeds run twice gave identical states. The code has 108 unit tests plus 12 for the web handler, and all pass, and a separate test runs each of the 17 port yards for four simulated hours while checking the safety rules.

G. Security

On 15 October 2026 the live site scored A+ on the Security Headers scan and A+ on all six server addresses in the SSL Labs test. Our own script passes 31 of 31 checks covering

TABLE V
VALUE OF PICKUP TIME INFORMATION (20 IMPORTS, 32 SEEDS)

What the yard knows	Rehandles	Truck wait (min)
Guess only	4.1 ± 0.5	9.8 ± 3.1
Estimate, 60 min off	3.1 ± 0.4	8.5 ± 2.3
Estimate, 15 min off	1.7 ± 0.4	7.2 ± 1.9
Booked 20 min ahead	4.0 ± 0.5	0.0 ± 0.0
Booked 1 hour ahead	4.0 ± 0.5	0.0 ± 0.0
Booked 2 hours ahead	3.8 ± 0.4	0.1 ± 0.1
Booked at arrival	1.3 ± 0.4	0.0 ± 0.0

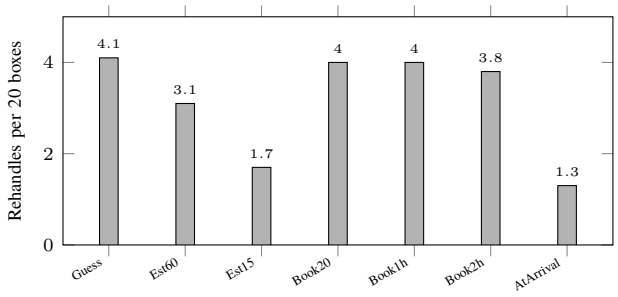


Fig. 4. Rehandles by what the yard knows. Only information known at placement helps.

policy headers, secure transport, blocked path tricks and file-type handling. We do not claim the site is unbreakable, only that it holds no real port data.

H. What it could save

The booking desk and an impact calculator turn these results into money, using a terminal's published internal shifting charge of INR 1,278 per move as a stand-in for the value of one move. As an example only, take VOC Port at 734,274 TEU a year and 1.5 TEU per box, which gives about 489,500 boxes. The port's real rehandle rate is not public, so we assume 10%, or about 48,950 rehandles. A 31% cut would avoid roughly 15,170 moves, worth about INR 1.94 crore a year. This is a what-if and not a measurement.

I. Limits

All results come from a simulator and a small yard. VOC Port's real layout and rehandle rate are not public, which is why the other port yards are invented. Time is compressed, so dwell in the model is much shorter than in real life. The model's vehicle both carries and stacks, which is closest to a straddle carrier. Our first-available baseline is weak, which is the reason we also report the 31% figure. And the hardware gate has not been built.

VI. SMART CITY RELEVANCE

Citizens and businesses. Shorter truck waits should mean less idling near the port and lighter traffic on the roads leading to it, and small businesses get the same booking desk as large ones. These benefits are what we expect from the simulation. We have not measured them at a real port.

Sustainability. Fewer wasted moves ought to mean less fuel or electricity per box, but we did not measure energy.

SDGs. The work connects to SDG 9 (industry, innovation and infrastructure), SDG 11 (sustainable cities), SDG 12 (responsible consumption and production) and SDG 13 (climate action).

Scale. The same engine runs yards from 12 to 60 positions, and the landing page shows 17 ports, so the idea can be demonstrated along a whole coast. A real roll-out would need actual yard data and links into national port systems.

VII. CONCLUSION AND FUTURE WORK

In simulation, Portlogi shows that a departure-aware allocator combined with safe vehicle control cuts rehandles, and that a booking desk helps most when the booking arrives at discharge. Next we plan to build and test the RFID gate, give the allocator forecasts of pickup time, and try the system on real yard data.

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